

Carlos Andrés Montoya Tobón

Solutions Architect & Senior Backend Engineer

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CORE STACK Java · Spring Boot · Microservices · AWS · Hexagonal Architecture · Keycloak · PostgreSQL · RabbitMQ

PROFESSIONAL SUMMARY

I'm Carlos Andrés Montoya Tobón, a Solutions Architect and Senior Backend Engineer based in Medellín, Colombia. For the last 18+ years I've been designing and building Java-based platforms — from transportation management systems and telecom transactional portals to fintech PaaS and blockchain credentials. My specialty is the architecture and development of web applications with Spring Framework, microservices and event-driven systems deployed on AWS, GCP and Azure. I've led distributed teams of up to 45 engineers and helped organizations modernize their stacks and delivery pipelines.

EXPERIENCE

► Solution Architect & Tech Lead

May 2026 — Present

Soldife · Remote

- Set the platform's architectural standard — hexagonal + DDD across 6 microservices and 9 SDKs, dependencies always pointing to the domain, enforced by ArchUnit on every build (zero architecture drift between PRs).
- Designed an ecosystem of 9 auto-configurable Spring Boot SDKs with contract governance — SemVer, backward compatibility and ADRs — published versioned to GitHub Packages.
- Established cross-cutting patterns: transactional outbox, canonical event envelope, end-to-end W3C tracing and centralized identity at the gateway.
- Documented architecture as code: C4 model (L1–L4) in Structurizr, ADRs and AI-oriented development guidelines.
- Designed CI/CD on GitHub Actions (GitFlow + shift-left) with reusable workflows parameterized by project type (Java, KrakenD, NGINX, contracts) and mandatory quality gates: build → tag-filtered unit/Karate tests → JaCoCo coverage → OWASP scan → ArchUnit → Spotless.
- Built SemVer-tagged base images on GHCR (Temurin 21, GraalVM native-image, KrakenD/Go toolchain) to cut per-job tool installs and speed pipelines; org-level governance via branch/PR rulesets and in-pipeline Slack notifications.

Java · Spring Boot · Hexagonal Architecture · DDD · Microservices · ArchUnit · GitHub Actions · KrakenD · GraalVM · Structurizr

► Backend Developer

Sep 2024 — Apr 2026

F2X · Flypass · Medellín, Colombia

- Designed a commissions-calculation component for the F2X memberships product (Java · Spring Boot · hexagonal architecture on AWS · Postgres).
- Designed a billing component for the F2X fuel-services product across the microservice fleet.
- Owned production-incident response and operational support for backend services.

Java · Spring Boot · Microservices · Hexagonal Architecture · AWS · PostgreSQL

Backend Developer

Dec 2023 — Dec 2024

Blerify · Remote

- Built security components with Keycloak and implemented KrakenD as the platform's API gateway.
- Developed a blockchain-based voucher/bond issuance component (Java · Vert.x · Kotlin · blockchain · Postgres on GCP/Kubernetes).

Java · Vert.x · Kotlin · Keycloak · KrakenD · Blockchain · PostgreSQL · GCP · Kubernetes

► Backend Developer

Apr 2022 — Jun 2023

Tul · Medellín, Colombia

- Built warehouse-management modules for the TUL B2B marketplace microservice fleet (Java · Spring Boot · hexagonal architecture on AWS · Postgres).
- Delivered Spring Security components and ran platform-wide code-quality reviews.

Java · Spring Boot · Spring Security · Microservices · Hexagonal Architecture · AWS · PostgreSQL

► Solutions Architect

Aug 2020 — Apr 2022

k4s · Klever for Solutions · Medellín, Colombia

- Architected PaaS solutions on AWS for MutualSER's next-gen health-insurance products (Colombian EPS).
- Designed reusable security and Keycloak/GCP integration frameworks consumed across product teams.
- Mentored developers on Spring, SOLID principles and clean architecture.

Java · Spring Boot · AWS · Keycloak · GCP

Solutions Architect

May 2016 — Apr 2020

Integral Software SAS · Medellín, Colombia

- Solutions Architect designing and evolving backend architectures on the Spring ecosystem over a four-year engagement.
- Designed multiple architectures and systems for the Mayor's Office of Medellín — among them Atenea and Tienda Virtual — applying hexagonal architecture, design patterns and security.
- Built components to migrate legacy systems to new architectures for Universidad de Medellín.
- Delivered the same legacy-to-modern architecture migration for the Colombian Air Force (FAC).

Kotlin · Java · Spring Boot · PostgreSQL · React · Angular · Keycloak · Hexagonal

Solutions Architect

Sep 2015 — May 2016

Software Estratégico · Medellín, Colombia

- Defined the new architectures for Matrix Tech's product line on Spring Boot, RabbitMQ and Cassandra.
- Built an SMS messaging gateway from scratch.
- Led the adoption of a DevOps culture to improve communication and collaboration across IT areas, enabling faster, more efficient delivery.

Java · Spring Boot · RabbitMQ · Cassandra · DevOps

► Technical Lead

Aug 2013 — Sep 2015

Tata Consultancy Services · Medellín, Colombia

- Led a distributed team of 45+ developers delivering Claro Perú's transactional self-service portal to production.
- Designed the Oracle Service Bus integration layer for Claro Colombia.
- Built WebSphere Portlet components in Java 8 and implemented CI infrastructure and Scrum practices.

Java · Oracle Service Bus · WebSphere · Jenkins

Lead Developer

Nov 2010 — Aug 2013

PATH S.A. · Medellín, Colombia

- Designed NetPlanet analysis components for the Claro network RCs and vehicle-tracking platforms.
- Built Mail Bomber and SMS-delivery systems in Java; IoT pilots for the city of Medellín.
- Designed AWS and on-premise infrastructure and CI/CD pipelines across the organization.

Java · AWS · IoT · CI/CD

Java Web Developer

Jun 2009 — Jun 2010

Ceiba Software House · Medellín, Colombia

- Built loan, policy and occupational-risk systems for Suramericana (Colombian insurance leader) in Java 7 / JEE 5.
- Authored Oracle SQL stored procedures and high-throughput backend components.

Java · JEE 5 · Oracle SQL

Java Web Developer

Apr 2007 — Jun 2009

Inversiones Informáticas Mercurio · Medellín, Colombia

- Built front-end and backend features for a Transportation Management System (TMS).
- Provided application support to 25+ companies running the system.

Java · JavaScript · JSP · SQL Server · MySQL

LIVE SITES

MyPipelines mypipelines.andresmontoyat.co — Reusable CI/CD pipeline library — parameterized GitHub Actions workflows (build, tag-filtered tests, JaCoCo coverage, OWASP scan, ArchUnit, Spotless) shared across JVM, KrakenD and NGINX projects.

MyGuidelines — Engineering Standards myguidelines.andresmontoyat.co — My engineering playbook, codified — the architecture, testing, security and clean-code standards I hold every project to, and the trade-offs behind them. Hexagonal + DDD for JVM microservices, written to be executable by engineers and AI coding agents.

TECHNICAL SKILLS

Backend

Java · **Kotlin** · Spring Boot · Spring Cloud · Spring Security · Hibernate · Kafka · REST / gRPC · Keycloak

Cloud & Infrastructure

AWS · GCP · Azure · **Kubernetes** · Docker · GKE · KrakenD · Terraform

DevOps & Tools

Jenkins · GitHub Actions · SonarQube · Nexus · Gradle · **Maven** · PostgreSQL · MySQL · Oracle SQL

AI & Productivity

Claude Code · GitHub Copilot · JetBrains Junie · Prompt Engineering